

FORT MCMURRAY, AB, Canada

WMO: 716890

Lat: 56.6500N

Lon: 111.2208W

Elev: 1211

StdP: 14.06

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.2	-27.2	-40.9	0.5	-32.0	-35.5	0.8	-26.9	20.8	23.1	18.0	19.5	2.2	100	0.663

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.4	84.0	63.3	80.4	61.7	77.2	60.2	66.0	78.7	64.1	75.6	62.4	73.2	8.7	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
61.7	86.1	69.6	59.8	80.2	68.0	58.0	75.1	66.3	31.4	79.3	30.0	75.8	28.7	73.1	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	16.6	14.4	-39.0	90.5	4.2	3.0	-42.1	92.7	-44.6	94.4	-47.0	96.1	-50.0	98.2
			-39.2	69.6	4.2	2.6	-42.2	71.4	-44.7	73.0	-47.1	74.4	-50.1	76.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	34.4	1.7	7.7	20.1	36.9	49.9	59.0	63.4	60.1	50.4	37.2	18.6	5.5	
	DBStd	24.59	16.92	14.68	15.34	10.59	8.76	6.29	5.19	6.32	8.06	8.78	12.35	15.67	
	HDD50	6951	1496	1183	928	403	109	7	0	5	92	407	941	1380	
	HDD65	11273	1961	1603	1393	842	472	197	91	173	442	863	1391	1845	
	CDD50	1239	0	0	0	11	105	277	416	319	102	9	0	0	
	CDD65	88	0	0	0	0	4	17	43	22	2	0	0	0	
	CDH74	1485	0	0	0	5	108	320	586	384	79	3	0	0	
	CDH80	341	0	0	0	0	24	67	141	91	18	0	0	0	
Wind	WSAvg	6.4	5.7	5.9	6.8	7.5	7.2	6.5	6.1	6.0	6.4	6.6	6.2	5.7	
Precipitation	PrecAvg	18.7	0.8	0.6	0.7	0.9	1.6	2.9	3.3	2.8	2.1	1.2	1.0	0.9	
	PrecMax	28.5	1.6	2.0	1.8	2.0	4.3	7.4	5.8	7.0	5.2	3.5	2.2	1.7	
	PrecMin	9.5	0.2	0.1	0.2	0.0	0.4	1.1	0.7	0.3	0.5	0.0	0.3	0.2	
	PrecStd	3.9	0.4	0.4	0.4	0.5	0.9	1.4	1.3	1.6	1.2	0.8	0.5	0.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.5	45.2	55.1	73.0	83.5	86.1	88.0	87.5	82.7	69.9	49.6	42.4	
		MCWB	37.2	37.0	41.9	52.3	58.8	62.3	66.2	65.7	61.7	54.0	40.7	36.3	
	2%	DB	37.8	38.9	49.5	64.9	76.4	81.3	84.0	82.2	75.1	60.2	42.6	36.4	
		MCWB	33.8	33.1	38.9	47.6	55.5	60.0	64.7	63.7	59.1	48.3	36.2	32.7	
	5%	DB	32.4	33.5	45.0	59.6	72.1	77.5	80.4	78.2	69.9	55.1	38.1	31.8	
		MCWB	29.6	29.4	36.3	44.6	53.2	58.3	62.7	62.2	56.1	45.7	33.9	29.1	
	10%	DB	25.8	28.8	40.7	54.8	67.8	74.0	76.8	74.4	65.2	50.6	34.5	27.0	
		MCWB	24.3	26.3	34.1	42.1	51.0	57.0	61.5	60.5	54.1	43.1	31.9	25.2	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	38.2	37.5	42.6	52.6	60.7	65.6	69.0	68.7	63.3	54.8	41.5	36.8
			MCDB	42.2	44.3	53.7	71.5	76.8	80.3	80.7	82.7	77.8	66.6	47.5	41.1
2%		WB	34.0	33.6	39.4	48.1	57.5	62.5	66.4	65.6	60.2	49.5	37.3	33.0	
		MCDB	37.6	38.7	48.5	63.5	72.9	74.2	78.8	78.2	72.9	58.1	41.5	36.0	
5%		WB	29.7	29.7	36.9	45.3	54.8	60.6	64.7	63.7	57.5	46.4	34.4	29.3	
		MCDB	32.3	33.5	44.3	58.3	68.6	71.6	76.0	75.1	67.7	54.0	37.5	31.8	
10%		WB	24.2	26.1	34.4	42.7	52.2	59.0	63.0	61.7	55.0	43.5	31.9	25.1	
		MCDB	25.8	28.8	40.2	53.7	65.2	70.3	73.7	72.1	63.2	49.9	34.3	27.0	
Mean Daily Temperature Range		5% DB	MDBR	17.1	21.0	22.8	22.1	24.5	22.8	22.4	22.4	21.5	16.4	14.2	15.8
			MCDBR	22.7	25.9	25.5	30.5	31.6	30.4	29.3	30.0	30.4	25.7	18.6	20.2
	MCWBR		19.5	20.8	16.8	17.3	15.3	13.9	13.5	14.7	16.5	15.6	14.2	17.4	
	5% WB	MCDBR	22.2	25.1	24.0	29.1	28.0	23.5	24.9	26.7	27.2	24.0	16.8	19.9	
		MCWBR	19.2	20.5	16.2	17.1	14.2	12.1	12.9	14.2	15.4	15.2	13.4	17.4	
Clear-Sky Solar Irradiance	taub	0.239	0.261	0.282	0.325	0.342	0.365	0.377	0.355	0.319	0.288	0.261	0.227		
	taud	2.296	2.333	2.377	2.415	2.434	2.377	2.403	2.432	2.532	2.578	2.382	2.224		
	Ebn at Noon	218	260	283	284	285	278	272	271	268	251	207	188		
	Edh at Noon	15	22	28	31	33	35	34	31	24	18	14	12		
All-Sky Solar Radiation	RadAvg	218	533	1038	1485	1763	1783	1715	1415	972	508	217	138		
	RadStd	19	29	65	103	177	147	92	104	96	74	27	10		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	+2.83	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page